

BENJAMIN MILLER

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LinkedIn | GitHub | Portfolio Website

EDUCATION

Computer Science B.S. - University of California, San Diego (UCSD) Sept. 2023 – June 2027
GPA: 3.8/4.0, Jacobs School of Engineering, UC San Diego Scholar's Society
Regent Scholar (Merit-based, top 1.5% of class)

Relevant Coursework: Advanced Data Structures and Algorithms (**C++**), Software Engineering (**JavaScript**), Operating Systems, Embedded Programming (**C**, **ARM Assembly**), Software Tools and Techniques (**Linux**, **Shell Scripts**), Data Science (**Machine Learning**, **Pandas**), Object Oriented Design (**Java**)

EXPERIENCE

Software Engineer Intern Sept. 2025 - Dec. 2025, Jun. 2026 - Sept. 2026
Western Digital *Irvine, CA*

- Developed internal tool for checking server availability for integration testing via a **Python microservice** and a full stack application (**Flask**, **SQLite**, **JavaScript**), as seen by successful end to end live demonstrations to management.
- Leveraged **Agentic AI** tools and **MCP** (model context protocol) integrations to automate development workflows resulting in higher efficiency.
- Designed and implemented **C++ APIs** for hard drives in an **Agile** development environment, validated through comprehensive unit and integration testing.
- Strengthened embedded C++ error-handling paths in a production firmware codebase, improving clarity and reliability of system error states.
- Expanded and improved the **unit testing** suite, resulting in increased overall **QA validation** quality of the **CI** (continuous integration) pipeline and higher quality code.
- Contributed to the development of a firmware feature for next generation HAMR (Heat-Assisted Magnetic Recording) storage devices.

Software Developer Intern Apr. 2025 - Aug. 2025
Center for Applied Internet Data Analysis *San Diego, CA*

- Enhanced maintainability and usability using **JavaScript** and **Python**, as demonstrated by successful contributions to the websites' codebase using **Git** for version control in large-scale site environments.
- Created useful automations using **Python** in a **Linux** environment resulting in a 75% decrease on average in manual labor required for workflows, such as content updates across the center's project sites.

Resident Advisor Jul. 2024 - Aug. 2024
Jacobs School of Engineering *La Jolla, CA*

- Mentored students in the COSMOS engineering summer program's Video Game Programming and AI Design group, providing guidance, support, and advice throughout the program.

LEADERSHIP

Electronic Props Designer Mar. 2026 - Present
Star Wars Club at UCSD *La Jolla, CA*

- Designed embedded systems and **C++** firmware for props for film making and on-campus promotional events using a real time operating system and microcontrollers resulting in screen accurate prop recreations.

Software Engineering Lead & Vice President Jun. 2024 - Jun. 2026
Themed Entertainment Association at UCSD *La Jolla, CA*

- Led development of software systems for themed attractions (**Python**, **C**, **C++**) which enhanced attraction operations that were attended by 100+ guests campuswide.

PROJECTS

Taro The Talking Bird - Interactive Robotic Figure Dec. 2025
C/C++, Multi-Threading, Advanced Linux Sound Architecture **Demo Video** | **Project Page** | **Github**

- Implemented a real-time **multithreaded system** using **C++** achieving audio processing with <10ms latency for synchronized motion control.
- Integrated local **AI models** for offline speech recognition and synthesis, processing **100+ live conversations** with 16K+ social media engagement.

SKILLS

- Programming: **Python**, **C**, **C++**, **Java**, **JavaScript**, **TypeScript**, **SQL**
- Web/App Development: **Flask**, **FastAPI**, **React**, **Node.js**, **REST APIs**, **Tailwind CSS**, **HTML**, **Vite**, **Express**
- Machine Learning: **NumPy**, **PyTorch**, **Pandas**, **scikit-learn**, **OpenCV**
- Tools & DevOps: **Linux**, **Git**, **Docker**, **CI/CD**, **Bash**, **Agile/Scrum**, **Virtual Machines**